

Q3(a)

The First Law of Thermodynamics states that the internal energy U of a system depends on its state, and the increase in internal energy ΔU is equal to sum of the heat supplied to the system Q and work done W on the system; i.e. $\Delta U = Q + W$.

Q3(b)

From the equation $p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$, the total kinetic energy or internal energy U of an ideal gas can be expressed as

$$K = \frac{1}{2} Nm \langle c^2 \rangle = \frac{3}{2} pV$$

From ideal gas equation, $pV = NkT$.

$$\text{Hence } U = \frac{3}{2} NkT$$

Q3(c)(i)1.

$$U = \frac{3}{2} NkT = \frac{3}{2} PV$$

$$\Delta U = \frac{3}{2} P_C V_C - \frac{3}{2} P_B V_B$$

$$\frac{3}{2} (75.5 \times 10^5) (4.54 - 2.27) \times 10^{-5}$$

$$= 257.08 \text{ J}$$

$$\approx 257 \text{ J}$$

Q3(c)(i)2.

For a constant pressure process, work done W by the gas in expansion $= p\Delta V$.

Hence work done on the gas $= -p\Delta V$

$$= - (75.5 \times 10^5) (4.54 - 2.27) \times 10^{-5}$$

$$= \mathbf{-171 \text{ J}}$$

Q3(c)(ii)

Stage of cycle	heat supplied to gas / J	work done on gas / J	increase in internal energy of the system / J
A $\rightarrow$ B	0	182	182 *
B $\rightarrow$ C	428 *	-171 *	257
C $\rightarrow$ D	0	-316	-316 *
D $\rightarrow$ A	-123 *	0 *	-123 *

Q4(a)(i)